What can waves tell us?

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Distinguish a signal's travel from the motion of its source.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 5, Day 10.*
The larger sphere has 1/3 of the smaller sphere's area per mass.

02. Worked example: explain the model
Pulse advances; marked segment moves locally and returns in the ideal model.

03. Read and interpret the model
No; the disturbance propagates while material oscillates locally in this model.

04. Apply the physics
2 s.

05. Explain your reasoning
Light travel delay; measurement uncertainty and motion during signal travel also matter.

06. Review: connect an earlier idea *Review: Unit 5, Day 1.*
Temperature change describes a thermal-state change (°C or K); heat is transferred energy (J). They are not the same quantity.

07. Exit check
Distance needs signal speed/time; velocity from Doppler needs wavelength/reference geometry; other justified pair.

08. STRETCH: test the model
Mark a rope segment or floating marker; compare its local motion with the advancing disturbance, using appropriate apparatus.

What can waves tell us?

antocciphysics | TEACHER KEY | HONORS + MATCHING SEI

Learning target: Distinguish a signal's travel from the motion of its source.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 5, Day 10.*
Area factor 9, mass factor 27; A/m becomes 1/3. Survival also depends on transfer rates, fragmentation and conditions, not only energy per mass.

02. Worked example: explain the model
Pulse advances; marked segment moves locally and returns in the ideal model.

03. Read and interpret the model
No; the disturbance propagates while material oscillates locally in this model.

04. Apply the physics
3.0×10^{6} m; signal covers out-and-back distance.

05. Explain your reasoning
Light travel delay; measurement uncertainty and motion during signal travel also matter.

06. Review: connect an earlier idea *Review: Unit 5, Day 1.*
No. Internal energy depends on amount of material and microscopic/material/phase structure as well as temperature.

07. Exit check
Second range 1.6 times first; delay does not establish mass.

08. STRETCH: test the model
Central 333.3 m/s; lower 300/0.95≈315.8, upper 300/0.85≈352.9 m/s.